Giải thuật Prim

```c
#include <stdio.h>
#define MAX_N 100
#define NO_EDGE -1
typedef struct {
    int A[MAX_N][MAX_N];
    int n, m;
}Graph;

void init_graph(Graph *G, int n) {
    G->n=n;
    G->m=0;
    int u, v;
    for (u=1;u<=n;u++)
        for (v=1;v<=n;v++)
            G->A[u][v] = NO_EDGE;
}

void add_edge(Graph *G, int u, int v, int w) {
    G->A[u][v] = w;
    G->A[v][u] = w;
    G->m++;
}
```

```c
#define oo 9999999
int pi[MAX_N], p[MAX_N], mark[MAX_N];

int Prim(Graph *G, Graph *T, int x) {
    int u, v, i;
    for (u=1;u<=G->n;u++) {
        mark[u] = 0;
        pi[u] = oo;
        p[u] = -1;
    }

    pi[x] = 0;

    for (i=1;i<G->n;i++) {
        int min_dist = oo;
        for (v=1;v<=G->n;v++)
            if (mark[v]==0 && pi[v]<min_dist) {
                min_dist = pi[v];
                u = v;
            }

        mark[u] = 1;

        for (v=1;v<=G->n;v++)
            if (G->A[u][v]!=NO_EDGE && mark[v]==0)
                if (G->A[u][v] < pi[v]) {
                    pi[v] = G->A[u][v];
                    p[v] = u;
                }
    }

    init_graph(T, G->n);
    int sum_w = 0;
    for (u=1;u<=G->n;u++)
        if (p[u]!=-1) {
            add_edge(T, p[u], u, G->A[p[u]][u]);
            sum_w += G->A[p[u]][u];
        }

    return sum_w;
}
```

```c
int main() {
    Graph G, T;
    int n, m, u, v, w, e;
    FILE *f = fopen("Kruscal_dt_130.txt", "r");
//  FILE *f = fopen("Kruscal_dt_163.txt", "r");
    fscanf(f,"%d%d", &n, &m);
    init_graph(&G, n);

    for (e=1;e<=m;e++) {
        fscanf(f,"%d%d%d", &u, &v, &w);
        add_edge(&G, u, v, w);
    }

    int sum = Prim(&G, &T, 1);
    printf("Cay khung co trong so nho nhat la %d", sum);
    for (u=1;u<=T.n;u++)
        for (v=u+1;v<=T.n;v++)
            if (T.A[u][v]!=NO_EDGE)
                printf("\n%d %d %d", u, v, T.A[u][v]);

    fclose(f);
    return 0;
}
```